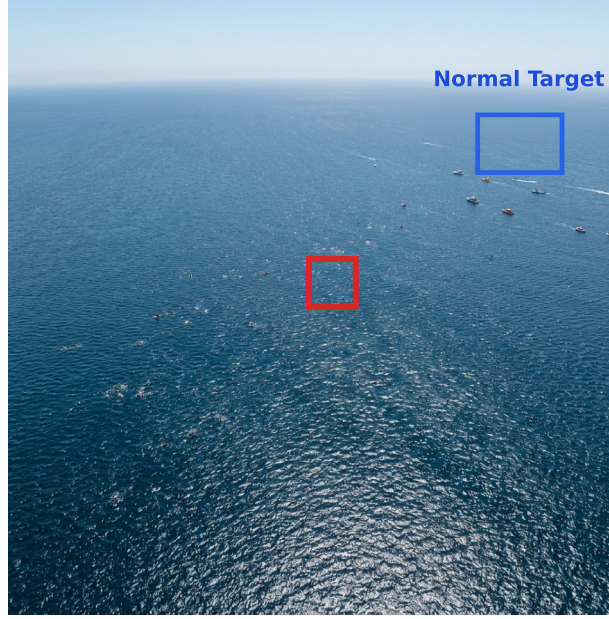


Input Image Tile (b1-tiled)

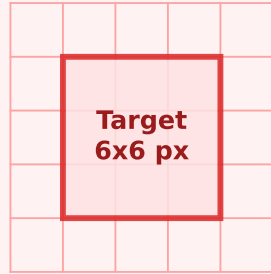
TinyPerson 800 x 800 px



Microscopic Target Callout

Spatial Discretization Noise Zone

- Target Scale:** $s \leq 8$ px
- Severe IoU Cliff Collapse
 - $\nabla_{\theta} \mathcal{L}_{\text{IoU}} = \mathbf{0}$ at $d \geq 6\text{px}$



ResNet-50 Backbone

Bottom-Up Shared Feature Extractor

C5 Stage

stride 32 | H/32 x W/32 x 2048



C4 Stage

stride 16 | H/16 x W/16 x 1024



C3 Stage

stride 8 | H/8 x W/8 x 512



C2 Stage

stride 4 | H/4 x W/4 x 256

Feature Pyramid Network

Top-Down Multi-Scale Semantic Pyramids

P5 Level

stride 32 | 256-d (Large Objects)



P4 Level

stride 16 | 256-d (Medium Objects)



P3 Level

stride 8 | 256-d (Small Objects)



P2 Level

stride 4 | 256-d (Highest Res)

PC-MOC Multi-Scale Cosine Feature Distillation

Prevents multi-scale representation drift during curriculum scale updates

$$\mathcal{L}_{\text{distill}} = \sum_{\ell=2}^5 \frac{1}{H_{\ell} W_{\ell}} \sum_{i,j} \left(1 - \frac{\langle f_{\text{curr}}^{P_{\ell}}(i,j), f_{\text{ref}}^{P_{\ell}}(i,j) \rangle}{\|f_{\text{curr}}^{P_{\ell}}(i,j)\| \|f_{\text{ref}}^{P_{\ell}}(i,j)\| + \varepsilon} \right)$$

- $f_{\text{curr}}^{P_{\ell}}$: Active student FPN features
- $f_{\text{ref}}^{P_{\ell}}$: Teacher reference representation

RoI Head with SA-ALW & Iterative Curriculum Routing

Scale-Adaptive Anisotropic Log-Wasserstein distance with scale-conditioned schedules

$$\mathcal{L}_{\text{reg}}^{\text{SA}} = \sqrt{w_{\text{pos}}(s) D_{\text{pos}}(p, t) + D_{\text{shape}}(p, t)} \quad K_{\text{SA}} = \exp \left[-\beta(s) \sqrt{D_{\text{SA}}(p, t)} \right]$$

- Adaptive temperature: $\beta(s) = \beta_{\min} + \Delta\beta \cdot u(s)$
- Position weight: $w_{\text{pos}}(s) = w_{\min} + \Delta w \cdot u(s)$

PC-MR Proposal Micro-Rescue with Orthogonal Gradient Projection

Eliminates gradient cancellation between dominant anchors and microscopic proposals

$$\mathbf{g}_{\text{proj}} = \mathbf{g}_{\text{micro}} - \frac{\mathbf{g}_{\text{micro}} \cdot \mathbf{g}_{\text{main}}}{\|\mathbf{g}_{\text{main}}\|^2 + \varepsilon} \mathbf{g}_{\text{main}} \quad (\mathbf{g}_{\text{proj}} \perp \mathbf{g}_{\text{main}})$$

- $\mathbf{g}_{\text{total}} = \mathbf{g}_{\text{main}} + \lambda_{\text{MR}} \mathbf{g}_{\text{proj}}$ guarantees zero gradient annihilation on microscopic targets